NUCLEAR FUSION AND ITER

NUCLEAR-PROCESS FIREWALL → D-T ENERGY MAP → FUSION-CONDITION TRIANGLE → CONFINEMENT COMPARISON → BREAKEVEN LADDER → NEUTRON-TO-MATERIAL CHAIN

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 NUCLEAR-PROCESS FIREWALL

NUCLEAR-PROCESS FIREWALL · SPLITS HEAVY NUCLEI AND POWERS CURRENT REACTORS · COMBINES LIGHT NUCLEI IN EXPERIMENTAL SYSTEMS · COMMERCIAL STATUS · FISSION OPERATING, FUSION NOT DEPLOYED · SHARED WORD NUCLEAR · DOES NOT ERASE PROCESS DIFFERENCES · TOPIC 05 CANNOT SUBSTITUTE FOR TOPIC 04

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
FISSION → splits heavy nuclei and powers current reactors	COMMERCIAL STATUS → fission operating, fusion not deployed	RULE → Topic 05 cannot substitute for Topic 04
FUSION → combines light nuclei in experimental systems	SHARED WORD NUCLEAR → does not erase process differences	FISSION → splits heavy nuclei and powers current reactors

SYSTEM / DEVICE / INSTITUTION
- FISSION → splits heavy nuclei and powers current reactors
- FUSION → combines light nuclei in experimental systems

DECISIVE TECHNICAL DISTINCTION
- COMMERCIAL STATUS → fission operating, fusion not deployed
- SHARED WORD NUCLEAR → does not erase process differences

STATUS / UNIT / EVIDENCE LIMIT
- RULE → Topic 05 cannot substitute for Topic 04
- FISSION → splits heavy nuclei and powers current reactors

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → Topic 05 cannot substitute for Topic 04.

01

STAGE 01 D-T ENERGY MAP

D-T ENERGY MAP · DEUTERIUM + TRITIUM · FUSION REACTION · CHARGED PRODUCT RETAINED FOR SELF-HEATING · CARRIES MOST ENERGY OUT OF PLASMA · CAPTURES HEAT AND SUPPORTS FUEL FUNCTIONS · PARTICLE ROLE CONTROLS ENGINEERING CONSEQUENCE

DEUTERIUM + TRITIUM → fusion reaction

HELIUM-4 / ALPHA → charged product retained for self-heating → NEUTRON → carries most energy out of plasma → BLANKET → captures heat and supports fuel functions

STARTING CONDITION / INPUT
- DEUTERIUM + TRITIUM → fusion reaction
- HELIUM-4 / ALPHA → charged product retained for self-heating

SCIENTIFIC OR ENGINEERING MECHANISM
- NEUTRON → carries most energy out of plasma
- BLANKET → captures heat and supports fuel functions

OUTPUT / FAILURE BOUNDARY
- RULE → particle role controls engineering consequence
- DEUTERIUM + TRITIUM → fusion reaction

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLANKET → captures heat and supports fuel functions.

02

STAGE 02 FUSION-CONDITION TRIANGLE

FUSION-CONDITION TRIANGLE · RAISES PROBABILITY OF OVERCOMING REPULSION · SUPPLIES REACTING PARTICLES · CONFINEMENT TIME · RETAINS ENERGY LONG ENOUGH · LAWSON TRIPLE PRODUCT · ALL THREE MUST COMBINE · A TEMPERATURE RECORD ALONE PROVES LITTLE

TEMPERATURE → raises probability of overcoming repulsion

DENSITY → supplies reacting particles → CONFINEMENT TIME → retains energy long enough → LAWSON TRIPLE PRODUCT → all three must combine

CONCEPT / ARCHITECTURE
- TEMPERATURE → raises probability of overcoming repulsion
- DENSITY → supplies reacting particles

MECHANISM / APPLICATION
- CONFINEMENT TIME → retains energy long enough
- LAWSON TRIPLE PRODUCT → all three must combine

READINESS / STATUS LIMIT
- TRAP → a temperature record alone proves little
- TEMPERATURE → raises probability of overcoming repulsion

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → a temperature record alone proves little.

03

STAGE 03 CONFINEMENT COMPARISON

CONFINEMENT COMPARISON · LOWER DENSITY AND LONGER CONFINEMENT · TORUS WITH STRONG PLASMA CURRENT · EXTERNALLY SHAPED STEADY-FIELD GEOMETRY · DENSE TARGET AND EXTREMELY SHORT CONFINEMENT · APPROACH AND DEVICE NAME ARE NOT POWER STATUS

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT

CONCEPT / ARCHITECTURE	MECHANISM / APPLICATION	READINESS / STATUS LIMIT
• TEMPERATURE → raises probability of overcoming repulsion • DENSITY → supplies reacting particles	• CONFINEMENT TIME → retains energy long enough • LAWSON TRIPLE PRODUCT → all three must combine	• TRAP → a temperature record alone proves little • TEMPERATURE → raises probability of overcoming repulsion

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → a temperature record alone proves little.

03

STAGE 03 CONFINEMENT COMPARISON

CONFINEMENT COMPARISON · LOWER DENSITY AND LONGER CONFINEMENT · TORUS WITH STRONG PLASMA CURRENT · EXTERNALLY SHAPED STEADY-FIELD GEOMETRY · DENSE TARGET AND EXTREMELY SHORT CONFINEMENT · APPROACH AND DEVICE NAME ARE NOT POWER STATUS

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
MAGNETIC → lower density and longer confinement	STELLARATOR → externally shaped steady-field geometry	RULE → approach and device name are not power status
TOKAMAK → torus with strong plasma current	INERTIAL → dense target and extremely short confinement	MAGNETIC → lower density and longer confinement

SYSTEM / DEVICE / INSTITUTION

- MAGNETIC → lower density and longer confinement
- TOKAMAK → torus with strong plasma current

DECISIVE TECHNICAL DISTINCTION

- STELLARATOR → externally shaped steady-field geometry
- INERTIAL → dense target and extremely short confinement

STATUS / UNIT / EVIDENCE LIMIT

- RULE → approach and device name are not power status
- MAGNETIC → lower density and longer confinement

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → approach and device name are not power status.

04

STAGE 04 BREAKEVEN LADDER

BREAKEVEN LADDER · PLASMA Q · FUSION POWER DIVIDED BY INJECTED HEATING · Q ABOVE ONE · PLASMA-LEVEL SCIENTIFIC THRESHOLD · ENGINEERING Q · INCLUDES RECIRCULATING PLANT LOADS · NET ELECTRICITY

PLASMA Q → fusion power divided by injected heating

Q ABOVE ONE → plasma-level scientific threshold → ENGINEERING Q → includes recirculating plant loads → NET ELECTRICITY → includes thermal conversion and all consumption

STARTING CONDITION / INPUT	SCIENTIFIC OR ENGINEERING MECHANISM	OUTPUT / FAILURE BOUNDARY
• PLASMA Q → fusion power divided by injected heating • Q ABOVE ONE → plasma-level scientific threshold	• ENGINEERING Q → includes recirculating plant loads • NET ELECTRICITY → includes thermal conversion and all consumption	• COMMERCIALITY → adds availability, cost and licensing • PLASMA Q → fusion power divided by injected heating

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NET ELECTRICITY → includes thermal conversion and all consumption.

05

STAGE 05 NEUTRON-TO-MATERIAL CHAIN

NEUTRON-TO-MATERIAL CHAIN · 14.1 MEV NEUTRON · ESCAPES MAGNETIC CONFINEMENT · RECEIVES ENERGY AND MAY BREED TRITIUM · DISPLACEMENT DAMAGE AND TRANSMUTATION · RADIOACTIVE MATERIAL-MANAGEMENT ISSUE · SEPARATE ENGINEERING FEASIBILITY CONSTRAINT

14.1 MeV NEUTRON → escapes magnetic confinement → BLANKET → receives energy and may breed tritium → STRUCTURE → displacement damage and transmutation → ACTIVATION → radioactive material-management issue → LIFETIME → separate engineering feasibility constraint

STARTING CONDITION / INPUT

- 14.1 MeV NEUTRON → escapes magnetic confinement
- BLANKET → receives energy and may breed tritium

SCIENTIFIC OR ENGINEERING MECHANISM

- STRUCTURE → displacement damage and transmutation
- ACTIVATION → radioactive material-management issue

OUTPUT / FAILURE BOUNDARY

- LIFETIME → separate engineering feasibility constraint
- 14.1 MeV NEUTRON → escapes magnetic confinement

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLANKET → receives energy and may breed tritium.

06

STAGE 06 TRITIUM FUEL LOOP

TRITIUM FUEL LOOP · D-T PLASMA · CONSUMES TRITIUM · FUSION NEUTRON · ENTERS LITHIUM-BEARING BLANKET · LITHIUM REACTION · PRODUCES NEW TRITIUM · RETURNS USABLE FUEL

D-T PLASMA → consumes tritium → FUSION NEUTRON → enters lithium-bearing blanket → LITHIUM REACTION → produces new tritium → EXTRACTION / PROCESSING → returns usable fuel → RULE → a test blanket is not a closed commercial cycle

STARTING CONDITION / INPUT

- D-T PLASMA → consumes tritium
- FUSION NEUTRON → enters lithium-bearing blanket

SCIENTIFIC OR ENGINEERING MECHANISM

- LITHIUM REACTION → produces new tritium
- EXTRACTION / PROCESSING → returns usable fuel

OUTPUT / FAILURE BOUNDARY

- RULE → a test blanket is not a closed commercial cycle
- D-T PLASMA → consumes tritium

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BLANKET → receives energy and may breed tritium.

06

STAGE 06 TRITIUM FUEL LOOP

TRITIUM FUEL LOOP · D-T PLASMA · CONSUMES TRITIUM · FUSION NEUTRON · ENTERS LITHIUM-BEARING BLANKET · LITHIUM REACTION · PRODUCES NEW TRITIUM · RETURNS USABLE FUEL

D-T PLASMA → consumes tritium → FUSION NEUTRON → enters lithium-bearing blanket → LITHIUM REACTION → produces new tritium → EXTRACTION / PROCESSING → returns usable fuel → RULE → a test blanket is not a closed commercial cycle

STARTING CONDITION / INPUT

- D-T PLASMA → consumes tritium
- FUSION NEUTRON → enters lithium-bearing blanket

SCIENTIFIC OR ENGINEERING MECHANISM

- LITHIUM REACTION → produces new tritium
- EXTRACTION / PROCESSING → returns usable fuel

OUTPUT / FAILURE BOUNDARY

- RULE → a test blanket is not a closed commercial cycle
- D-T PLASMA → consumes tritium

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → a test blanket is not a closed commercial cycle.

07

STAGE 07 PLASMA EXHAUST PATH

PLASMA EXHAUST PATH · CORE PLASMA · CREATES HEAT AND HELIUM ASH · EDGE TRANSPORT · MOVES PARTICLES AND HEAT OUTWARD · EXHAUSTS IMPURITIES AND HELIUM · PLASMA-FACING MATERIAL · RECEIVES EXTREME HEAT LOAD

CORE PLASMA → creates heat and helium ash → EDGE TRANSPORT → moves particles and heat outward → DIVERTOR → exhausts impurities and helium → PLASMA-FACING MATERIAL → receives extreme heat load → RULE → confinement success does not solve exhaust lifetime

STARTING CONDITION / INPUT

- CORE PLASMA → creates heat and helium ash
- EDGE TRANSPORT → moves particles and heat outward

SCIENTIFIC OR ENGINEERING MECHANISM

- DIVERTOR → exhausts impurities and helium
- PLASMA-FACING MATERIAL → receives extreme heat load

OUTPUT / FAILURE BOUNDARY

- RULE → confinement success does not solve exhaust lifetime
- CORE PLASMA → creates heat and helium ash

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → confinement success does not solve exhaust lifetime.

08

STAGE 08 ITER MISSION CARD

ITER MISSION CARD · UNDER CONSTRUCTION · EXPERIMENTAL INTERNATIONAL TOKAMAK · BURNING-PLASMA STUDY · CORE SCIENTIFIC PURPOSE · Q=10 DESIGN · 500 MW FUSION FROM 50 MW PLASMA HEATING · NO TURBINE

UNDER CONSTRUCTION → experimental international tokamak

BURNING-PLASMA STUDY → core scientific purpose → Q=10 DESIGN → 500 MW fusion from 50 MW plasma heating → NO TURBINE → no conversion to grid electricity

STARTING CONDITION / INPUT

- UNDER CONSTRUCTION → experimental international tokamak
- BURNING-PLASMA STUDY → core scientific purpose

SCIENTIFIC OR ENGINEERING MECHANISM

- Q=10 DESIGN → 500 MW fusion from 50 MW plasma heating
- NO TURBINE → no conversion to grid electricity

OUTPUT / FAILURE BOUNDARY

- TEST TECHNOLOGIES → bridge to later machines
- UNDER CONSTRUCTION → experimental international tokamak

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO TURBINE → no conversion to grid electricity.

09

STAGE 09 INDIA-IN-ITER MAP

INDIA-IN-ITER MAP · ONE OF SEVEN ITER MEMBERS · IN-KIND PACKAGES · SPECIFIED HARDWARE AND SYSTEMS · DAE-AIDED PLASMA-RESEARCH INSTITUTE · DOMESTIC PROCUREMENT AND DELIVERY AGENCY IN IPR · MEMBERSHIP, INSTITUTE, AGENCY AND PACKAGE DIFFER

INDIA → one of seven ITER members

IN-KIND PACKAGES → specified hardware and systems → IPR → DAE-aided plasma-research institute → ITER-INDIA → domestic procurement and delivery agency in IPR

CONCEPT / ARCHITECTURE

- INDIA → one of seven ITER members
- IN-KIND PACKAGES → specified hardware and systems

MECHANISM / APPLICATION

- IPR → DAE-aided plasma-research institute
- ITER-INDIA → domestic procurement and delivery agency in IPR

READINESS / STATUS LIMIT

- RULE → membership, institute, agency and package differ
- INDIA → one of seven ITER members

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → membership, institute, agency and package differ.

INDIA-IN-ITER MAP · ONE OF SEVEN ITER MEMBERS · IN-KIND PACKAGES · SPECIFIED HARDWARE AND SYSTEMS · DAE-AIDED PLASMA-RESEARCH INSTITUTE · DOMESTIC PROCUREMENT AND DELIVERY AGENCY IN IPR · MEMBERSHIP, INSTITUTE, AGENCY AND PACKAGE DIFFER

INDIA → one of seven ITER members

IN-KIND PACKAGES → specified hardware and systems → **IPR → DAE-aided plasma-research institute** → **ITER-INDIA → domestic procurement and delivery agency in IPR**

CONCEPT / ARCHITECTURE

- INDIA → one of seven ITER members
- IN-KIND PACKAGES → specified hardware and systems

MECHANISM / APPLICATION

- IPR → DAE-aided plasma-research institute
- ITER-INDIA → domestic procurement and delivery agency in IPR

READINESS / STATUS LIMIT

- RULE → membership, institute, agency and package differ
- INDIA → one of seven ITER members

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → membership, institute, agency and package differ.

10

STAGE 10 DOMESTIC RESEARCH DEVICES

DOMESTIC RESEARCH DEVICES · EXPERIMENTAL TOKAMAK AND PLASMA CAMPAIGNS · SUPERCONDUCTING LONG-PULSE RESEARCH PLATFORM · RESEARCH OUTPUT · DIAGNOSTICS, CONTROL AND OPERATING KNOWLEDGE · NOT OUTPUT · COMMERCIAL FUSION ELECTRICITY · EXPERIMENTAL OPERATION IS NOT POWER GENERATION

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
ADITYA-U → experimental tokamak and plasma campaigns	RESEARCH OUTPUT → diagnostics, control and operating knowledge	RULE → experimental operation is not power generation
SST-1 → superconducting long-pulse research platform	NOT OUTPUT → commercial fusion electricity	ADITYA-U → experimental tokamak and plasma campaigns

SYSTEM / DEVICE / INSTITUTION

- ADITYA-U → experimental tokamak and plasma campaigns
- SST-1 → superconducting long-pulse research platform

DECISIVE TECHNICAL DISTINCTION

- RESEARCH OUTPUT → diagnostics, control and operating knowledge
- NOT OUTPUT → commercial fusion electricity

STATUS / UNIT / EVIDENCE LIMIT

- RULE → experimental operation is not power generation
- ADITYA-U → experimental tokamak and plasma campaigns

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → experimental operation is not power generation.

11

STAGE 11 FUSION-READINESS ANSWER SPINE

FUSION-READINESS ANSWER SPINE · D-T FUSION PLASMA CONFINEMENT LAWSON AND Q · ALPHA HEATING NEUTRON BLANKET TRITIUM AND MATERIALS · PLASMA GAIN ENGINEERING GAIN AND NET ELECTRICITY · ITER INDIA IPR ITER-INDIA ADITYA-U SST-1 AND DEMO · DATED SCHEDULE COST CONTRIBUTION AND MILESTONE EVIDENCE

DEFINE → D-T fusion plasma confinement Lawson and Q → **TRACE → alpha heating neutron blanket tritium and materials** → **SEPARATE → plasma gain engineering gain and net electricity** → **MAP → ITER India IPR ITER-India ADITYA-U SST-1 and DEMO** → **VERIFY → dated schedule cost contribution and milestone evidence**

CONCEPT / ARCHITECTURE

- DEFINE → D-T fusion plasma confinement Lawson and Q
- TRACE → alpha heating neutron blanket tritium and materials

MECHANISM / APPLICATION

- SEPARATE → plasma gain engineering gain and net electricity
- MAP → ITER India IPR ITER-India ADITYA-U SST-1 and DEMO

READINESS / STATUS LIMIT

- VERIFY → dated schedule cost contribution and milestone evidence
- DEFINE → D-T fusion plasma confinement Lawson and Q

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERIFY → dated schedule cost contribution and milestone evidence.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · ITER ORGANIZATION · GLOBAL FLAGSHIP EXPERIMENTAL FUSION COLLABORATION OF SEVEN MEMBERS — · CHINA, JAPAN, SOUTH KOREA, RUSSIA AND THE USA — · INDIA AS ITER MEMBER · CONTRIBUTES NINE IN-KIND PACKAGES — CRYOSTAT, IN-WALL SHIELDING, COOLING-WATER · SYSTEM, ION-CYCLOTRON RF HEATING, ELECTRON-CYCLOTRON RF HEATING, DIAGNOSTIC NEUTRAL · INSTITUTE FOR PLASMA RESEARCH (IPR), GANDHINAGAR

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- ITER Organization: global flagship experimental fusion collaboration of seven members — India, EU,
- China, Japan, South Korea, Russia and the USA — sited at Saint-Paul-lez-Durance, France.
- India as ITER member: contributes nine in-kind packages — cryostat, in-wall shielding, cooling-water system, cryogenic
- system, ion-cyclotron RF heating, electron-cyclotron RF heating, diagnostic neutral beam, power supplies and diagnostics.

READINESS, UNIT OR STATUS LIMIT

- Institute for Plasma Research (IPR), Gandhinagar: a DAE-aided institute
- India's plasma-physics and fusion-research base, operating ADITYA-U and SST-1.
- ITER-India: the Indian Domestic Agency for ITER, constituted as a dedicated project within IPR .
- It contracts Indian industry, manages delivery schedules and interfaces with the ITER Organization.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- CAUTION: Writing "IPR / ITER-India" as one entity blurs a research institute with a procurement agency — precisely
- SST-1 and Aditya-U: domestic experimental platforms that cultivate operational knowledge and scientific manpower
- IPR describes them as research devices, not power reactors.
- CAUTION: Fusion cooperation is simultaneously scientific diplomacy, industrial capability-building and long-horizon energy strategy.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels